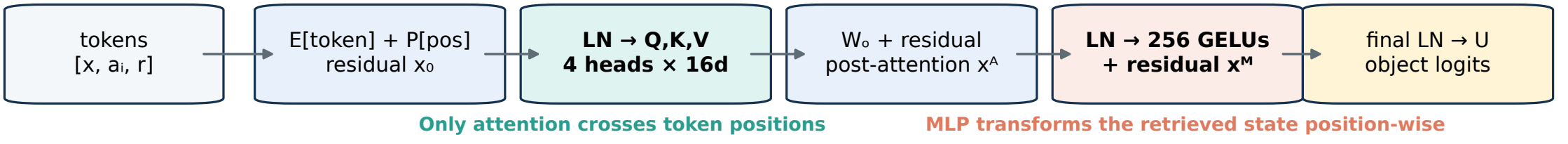
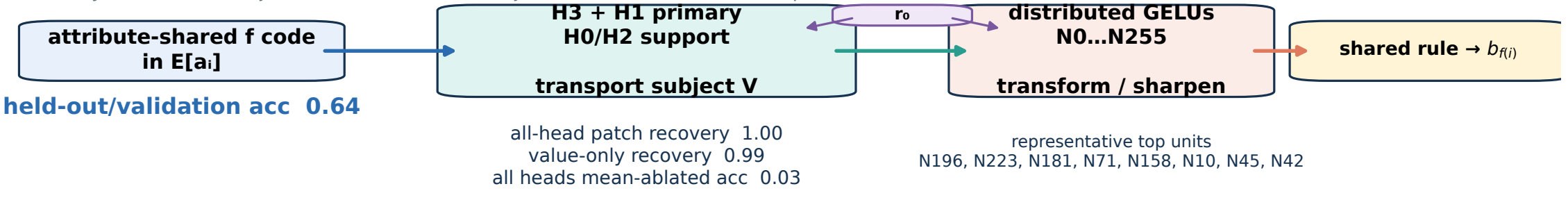
From generic transformer blocks to the compute units used for factual recall

A Literal forward pass



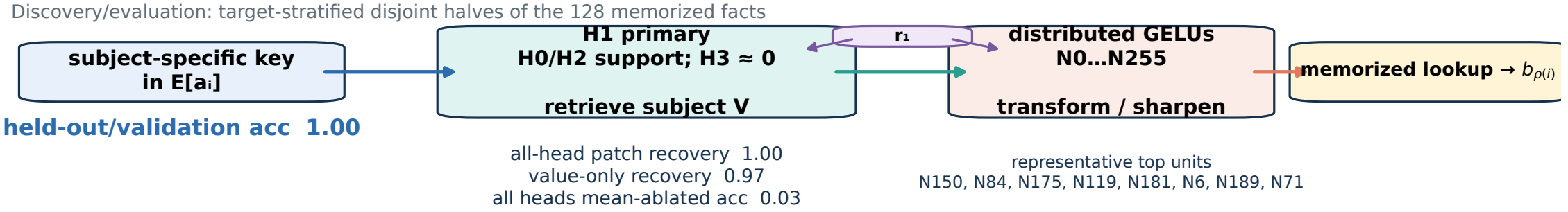
B Goal 1 compute graph — semantic, F=16

Discovery: 128 trained subjects → evaluation: 64 subjects never seen in relation triples

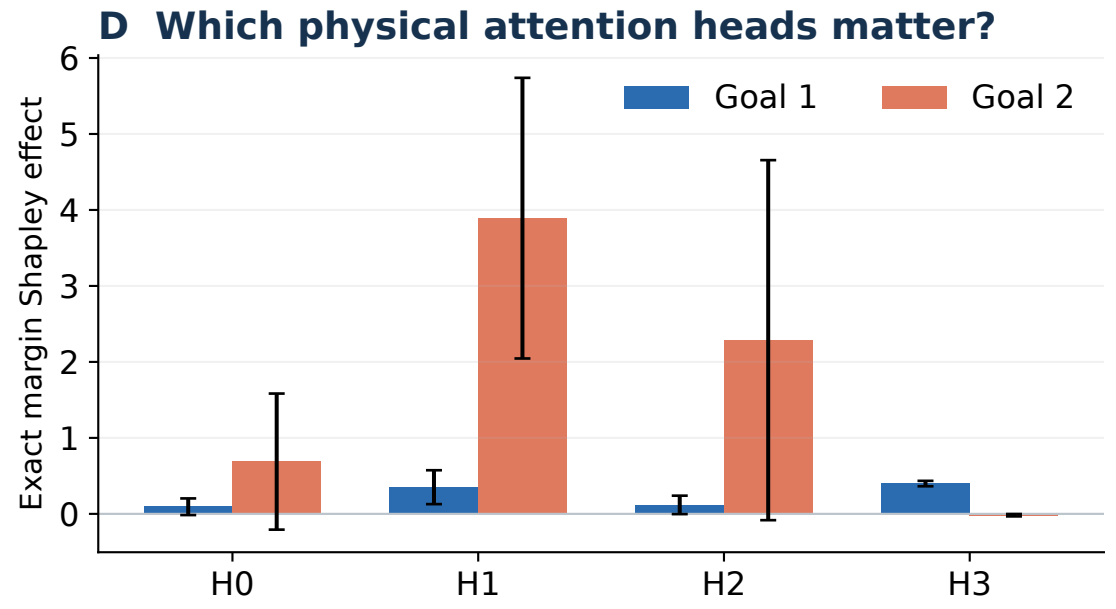


C Goal 2 compute graph — random, F=64

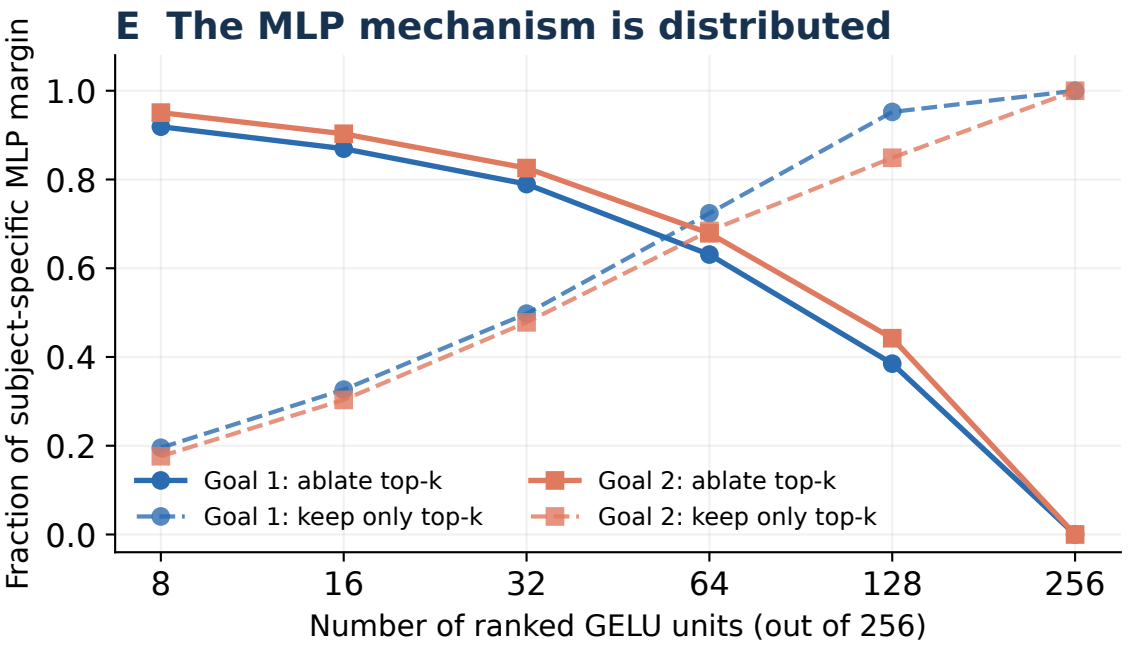
Discovery/evaluation: target-stratified disjoint halves of the 128 memorized facts



D Which physical attention heads matter?



E The MLP mechanism is distributed



Bars/curves: mean over seeds 19, 31, 47; error bars: across-seed SD. All causal rankings are evaluated on subjects disjoint from discovery.